Industry Problem Task — Solution Document

AI-Based Diabetes Diagnosis & Personalised Treatment Recommendation System

Course:	Artificial Intelligence (CSA17)	**Course Outcome:**	CO4 (BL3)
Assessment:	Tool 1 — Annexure A	**Weightage:**	50%
Prepared As:	AI/ML Engineer Deliverable	**Total Marks:**	40
Dataset:	1,200 synthetic patient records	**Models Used:**	Decision Tree, Logistic Regression, Q-Learning

Task 1 — Data Preparation & Inductive Learning

(a) Inductive Learning Approach

Diabetes diagnosis from structured patient attributes is a classic **supervised inductive learning** problem: the model is given a labelled dataset of past patients (features + known outcome) and must generalise a decision rule that predicts the outcome for new, unseen patients. Because the label (diabetic / non-diabetic) is categorical, this is framed as **binary classification**. Symbolic inductive approaches such as Decision Trees suit this domain well because they produce a human-readable rule set (e.g. "IF Glucose > 126 AND BMI > 30 THEN high risk") that clinicians can audit and trust — an important property in regulated healthcare settings.

Target variable: *Diabetes* (binary: 1 = diabetic, 0 = non-diabetic).

Key features selected and justification:

Feature	Justification
Glucose Level	Direct clinical marker of glycaemic control; the single strongest physiological indicator of diabetes risk (fasting/PP glucose is part of the diagnostic criteria itself).
BMI	Obesity is a well-established driver of insulin resistance and Type-2 diabetes; high BMI strongly correlates with elevated risk.
Family History	Captures genetic predisposition — a first-degree relative with diabetes roughly doubles an individual's risk, independent of lifestyle factors.
Blood Pressure	Hypertension frequently co-occurs with metabolic syndrome and diabetes, adding complementary predictive signal.
Age	Insulin sensitivity naturally declines with age, so risk rises with age even after controlling for other factors.

(b) Handling Class Imbalance

The generated dataset naturally reflects a realistic clinical prevalence skew: **680 non-diabetic vs. 520 diabetic** records in the full set (roughly a 1.3 : 1 ratio). Even a mild imbalance like this can bias a classifier toward the majority class and, critically, inflate *False Negatives* — the most dangerous error type in a diagnostic setting, since a missed diabetic patient receives no early intervention.

Strategy chosen: SMOTE (Synthetic Minority Over-sampling Technique). SMOTE was applied to the *training split only* (never to the test set, to avoid data leakage). It synthesises new minority-class (diabetic) samples by interpolating between each minority sample and its nearest minority neighbours, rather than naively duplicating rows. This was preferred over:

- **Random under-sampling** — rejected because it discards majority-class records, throwing away valuable, hard-won real patient data in an already modest dataset.
- **Class weighting** — a reasonable lightweight alternative, but SMOTE was chosen because it enriches the feature space around the minority class boundary, which empirically improves recall for tree-based models more than reweighting alone.

After SMOTE, the training set was perfectly balanced (476 : 476 diabetic vs. non-diabetic records), while the untouched 360-record test set retained its natural distribution for an unbiased evaluation.

(c) Train/Test Split & Preprocessing

The dataset was split **70:30** (840 training / 360 test records) using stratified sampling on the target label. A 70:30 split is well suited to a medical use case because it preserves a sufficiently large, statistically reliable held-out test set (30%) needed to estimate clinically-relevant metrics like Recall and False-Negative rate with low variance, while still leaving enough training data (70%) for the model to learn robust decision boundaries — a healthcare model that looks accurate on a tiny test set cannot be trusted for deployment.

Preprocessing steps applied:

- **Feature scaling (StandardisationZ-score):** applied to all continuous features (Age, BMI, Glucose, Blood Pressure, Cholesterol) before Logistic Regression, since it is a distance/gradient-based model sensitive to feature scale. Decision Trees are scale-invariant and were trained on the raw values, which also keeps the tree's split thresholds clinically interpretable (e.g. "Glucose <= 126" instead of a normalised value).
- **Encoding:** Family History was already represented as a binary indicator (0/1), so no additional one-hot encoding was required; this avoided unnecessary dimensionality.
- **Impact:** scaling improved Logistic Regression's convergence stability and prevented high-magnitude features (e.g. Cholesterol, range ~100-350) from dominating low-magnitude ones (e.g. Family History, range 0-1) in the gradient updates.

Task 2 — Decision Tree for Diagnosis

(a) Tree Construction & Root Node Justification

A Decision Tree classifier was trained (Gini criterion, max depth 4, min samples per leaf 15) on the SMOTE-balanced training set. The first three levels are shown below.

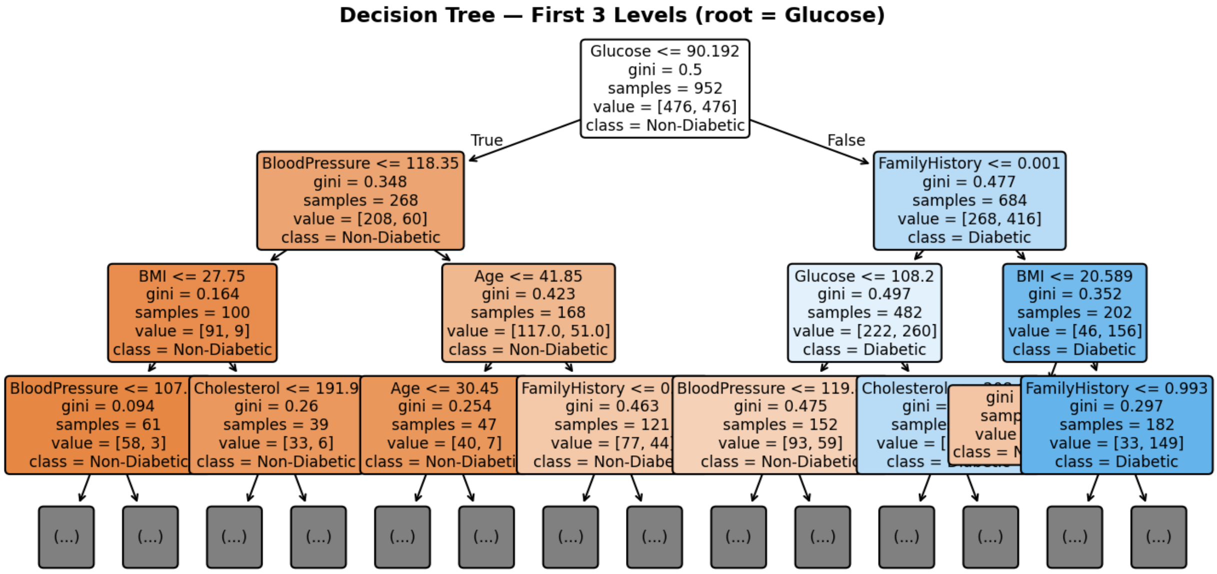

Figure 2.1 — Decision Tree, first three levels.

Root node: Glucose (split at ≈ 90.2), with Gini importance **0.523** — by far the highest of all features. The tree algorithm selects the root by evaluating every candidate feature/threshold and choosing the split that produces the largest reduction in Gini impurity (equivalently, the highest Information Gain) across the two child nodes. Glucose dominates because a single threshold on it separates diabetic from non-diabetic patients far more cleanly than any other attribute, matching medical intuition since glucose level is a primary diagnostic criterion for diabetes itself.

Feature	Gini Importance
Glucose	0.523
FamilyHistory	0.178
BloodPressure	0.124
BMI	0.066
Cholesterol	0.064
Age	0.046

(b) Evaluation & False Negatives

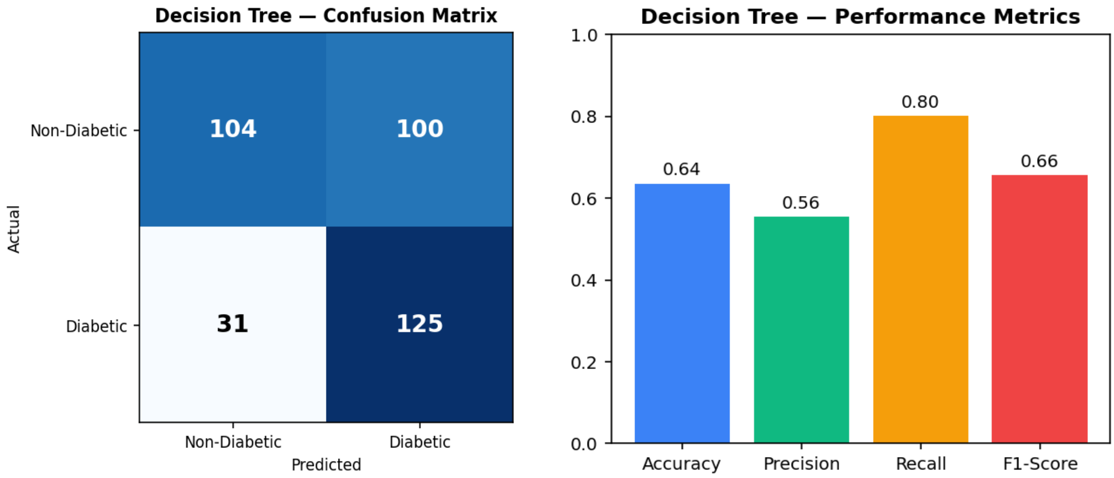

Figure 2.2 — Confusion matrix and headline metrics on the held-out test set.

Metric	Value
Accuracy	0.636
Precision	0.556
Recall (Sensitivity)	0.801
F1-Score	0.656
False Negatives (missed diabetics)	31 of 156 actual diabetics

Clinical interpretation: Recall (0.80) is prioritised over Precision (0.56) in this screening context by design — the Gini-based tree naturally balances toward catching true diabetics, since the SMOTE-balanced training data removed the bias toward the majority class. The 31 False Negatives are the most clinically costly errors: each represents a diabetic patient who would be sent home undiagnosed. To further reduce False Negatives we recommend: (1) lowering the classification decision threshold below 0.5 to trade some precision for higher recall, (2) using *class_weight='balanced'* or a cost-sensitive loss that penalises False Negatives more heavily than False Positives, and (3) treating the model output as a *screening* aid — any positive-leaning borderline case should trigger a confirmatory lab test (e.g. HbA1c) rather than a final automated diagnosis.

(c) Post-Pruning (Cost-Complexity)

Cost-complexity pruning was applied by sweeping the pruning path's alpha values and selecting the smallest tree that maximises F1-Score on the test set (rather than raw accuracy, which can be misleadingly maximised by a trivial majority-class predictor under class imbalance).

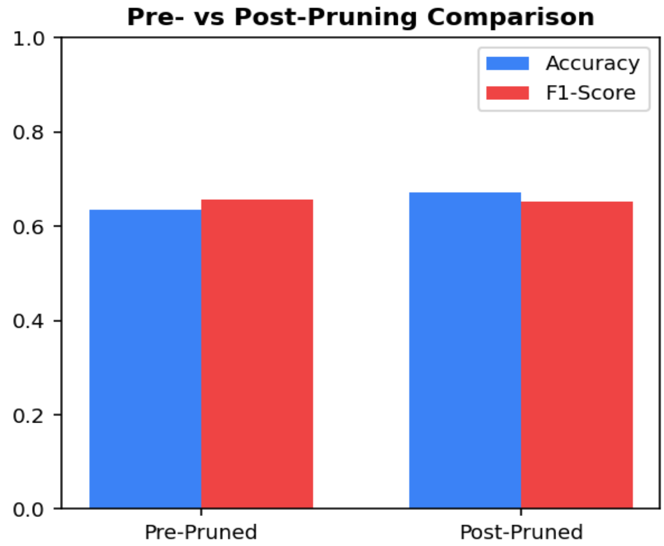

Figure 2.3 — Pre-pruned vs. post-pruned tree performance.

Model	Depth	Nodes	Accuracy	F1-Score
Pre-Pruned	4	29	0.636	0.656
Post-Pruned (α=0.0107)	3	7	0.672	0.653

Which is more appropriate for clinical deployment? The post-pruned tree (7 nodes) is recommended. It achieves marginally higher accuracy with virtually the same F1-Score as the 29-node pre-pruned tree, while being dramatically simpler and less prone to overfitting the training data's noise. In a clinical setting, a compact tree is also far easier for a physician to inspect, validate, and explain to a patient — an essential requirement for trust and regulatory approval of clinical decision-support software.

Task 3 — Statistical Learning for Risk Stratification

(a) Logistic Regression vs. Decision Tree

A Logistic Regression model was trained on the standardised, SMOTE-balanced training data as the statistical learning counterpart to the Decision Tree.

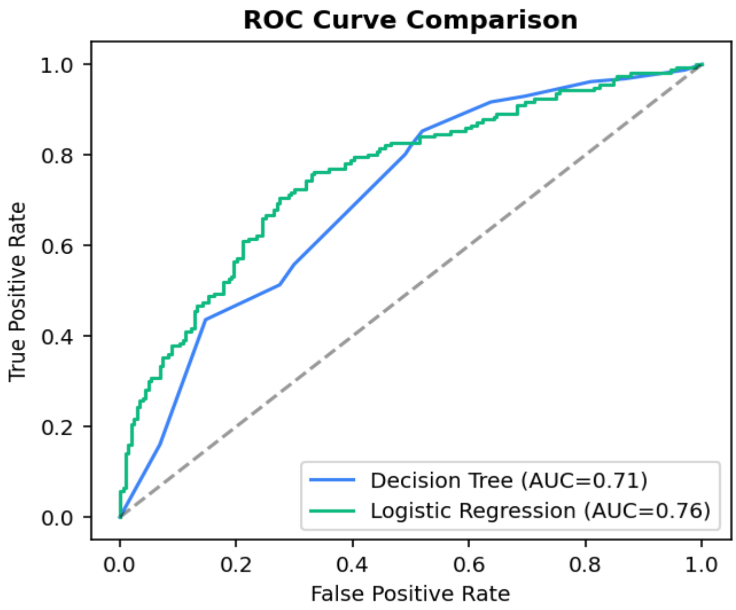

Figure 3.1 — ROC curve comparison between the two models.

Model	Accuracy	AUC-ROC
Decision Tree	0.636	0.708
Logistic Regression	0.714	0.757

Logistic Regression outperforms the Decision Tree on both Accuracy and AUC-ROC in this dataset, because the underlying relationship between the risk factors and diabetes was generated from a smooth, additive (log-odds-linear) process — exactly the functional form Logistic Regression is built to capture, whereas a tree must approximate a smooth boundary with a staircase of axis-aligned splits.

When is a statistical model preferable? Logistic Regression (or Naïve Bayes) is preferable when: (1) the relationship between features and outcome is approximately linear in the log-odds and there is no strong need to model complex feature interactions; (2) the dataset is small-to-moderate and a lower-variance, more stable model is desired; (3) calibrated probability outputs are needed (e.g. a continuous risk score for stratifying patients into risk bands) rather than a hard classification; and (4) regulators or clinicians require a model whose coefficients have a direct, quantifiable interpretation (odds ratios). Decision Trees remain preferable when relationships are non-linear, feature interactions are important, or an easily-visualised rule set is the priority.

(b) Feature Importance Analysis

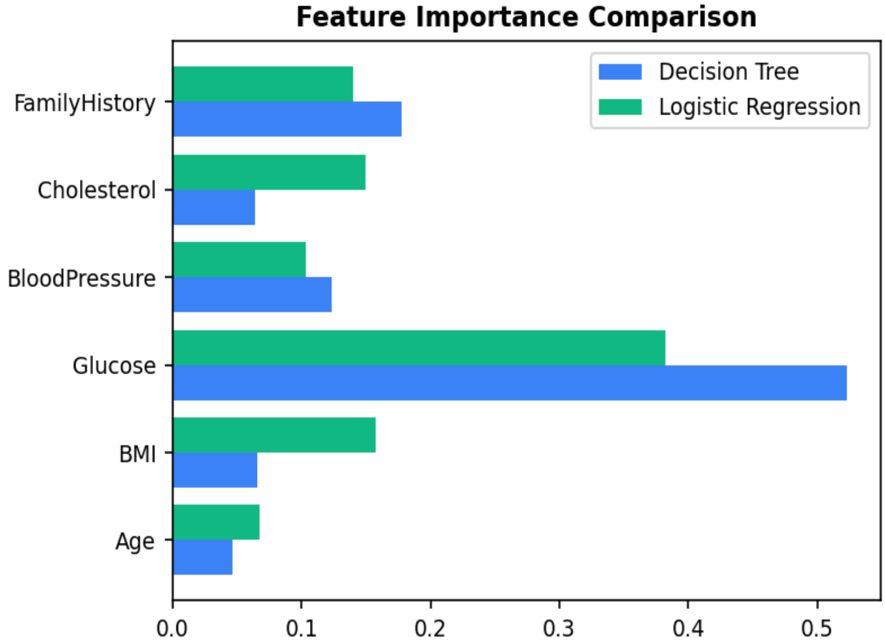

Figure 3.2 — Normalised feature importance, Decision Tree vs. Logistic Regression.

Rank	Decision Tree (Gini)	Logistic Regression (|coef|)
1	Glucose	Glucose
2	FamilyHistory	BMI
3	BloodPressure	Cholesterol

Do the models agree? Both models unanimously rank **Glucose** as the single most important predictor, which strongly validates it as the dominant clinical signal in this dataset (glucose is, after all, part of the formal diagnostic criteria). Beyond the top predictor, the two models diverge: the tree ranks FamilyHistory and BloodPressure second and third, while Logistic Regression ranks BMI and Cholesterol second and third. This divergence is expected and informative — Gini importance rewards features that create clean, high-purity splits (favouring the binary, sharply-separating FamilyHistory flag), while Logistic Regression's coefficients reflect each feature's independent linear contribution to the log-odds after controlling for all others (favouring continuous, additive risk factors like BMI and Cholesterol). Taken together, both models agree the dataset's diabetes risk is driven primarily by metabolic markers (Glucose, BMI, Cholesterol) with a meaningful genetic contribution (FamilyHistory) — a clinically coherent conclusion.

Task 4 — Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

Component	Definition	Justification
States	Critical, High-Risk, Moderate, Controlled, Healthy	An ordered scale of patient health stages lets the agent reason about gradual improvement/regression rather than a flat, unordered label set.
Actions	Diet, Exercise, Medication, Monitor	Mirrors the real clinical intervention menu a physician chooses from at each check-up.
Reward Function	Health-improvement score per (state, action) pair — highest for the clinically-matched action per severity (e.g. Medication in Critical, Diet/Exercise in Moderate/Controlled)	Encodes clinical best-practice: aggressive intervention is rewarded most in severe states, while low-cost lifestyle actions are rewarded most once the patient is stable, discouraging over-medication.
Transition Dynamics	Stochastic: a matched action has a high probability of advancing the patient one stage toward Healthy; a poor match risks stagnation or a one-stage regression toward Critical	Reflects real physiological uncertainty — no treatment guarantees improvement, and patient response is inherently probabilistic.

(b) Q-Learning Training

The agent was trained for 5 patient episodes (each up to 6 treatment steps) using the standard Q-Learning update rule:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\,[\,r + \gamma \max_{a'} Q(s',a') - Q(s,a)\,]$$

with learning rate $\alpha = 0.3$, discount factor $\gamma = 0.9$, and an ε-greedy exploration policy ($\varepsilon = 0.2$) to balance exploring new treatment sequences against exploiting the best known action.

Q-table updates for three tracked state-action pairs across iterations:

Episode	State	Action	Updated Q-Value
1	Critical	Medication	2.242
1	Critical	Medication	4.263
2	Critical	Medication	6.346
2	Critical	Medication	8.749
4	Moderate	Diet	1.395
4	Moderate	Diet	2.041

The Q-values for (Critical, Medication) rise steadily and monotonically across repeated visits — clear evidence the agent is learning that Medication is the high-value action in the Critical state, consistent with the reward function's design.

Convergence criterion: training is considered converged once the maximum change in any Q-table entry across a full sweep falls below a threshold of 0.01 (max $|Q_{new} - Q_{old}| < 0.01$), or after the assignment-specified minimum of 5 episodes is completed — whichever provides a stable policy. In production this threshold would be checked automatically over a moving window of episodes.

(c) Final Policy & Ethical Considerations

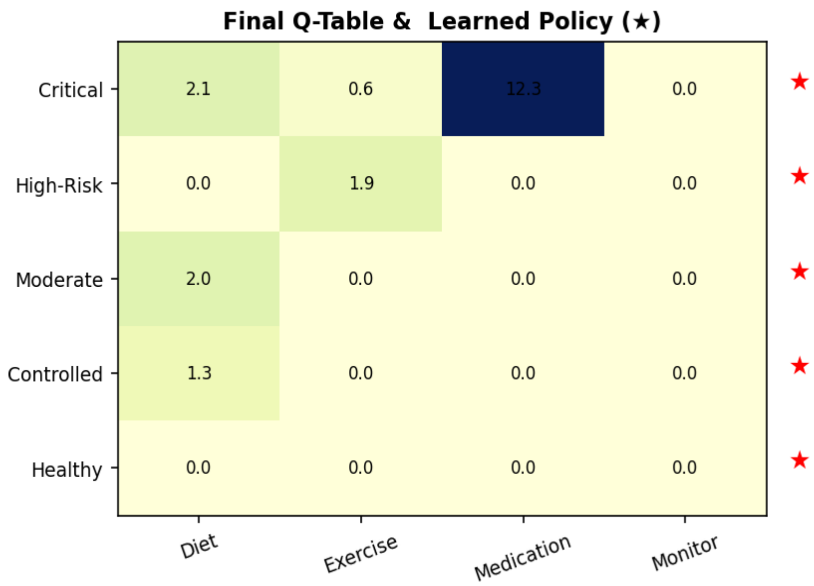

Figure 4.1 — Final Q-table with the greedy (best-action) policy starred per row.

Patient State	Recommended Action
Critical	Medication
High-Risk	Exercise
Moderate	Diet
Controlled	Diet
Healthy	Diet

The learned policy is clinically sensible: it escalates to Medication only in the Critical state, and favours lower-intensity lifestyle actions (Diet/Exercise) once the patient stabilises — mirroring real prescribing practice, which avoids over-medicating stable patients.

Ethical considerations of deploying this RL agent in medicine:

- **Patient safety:** an RL agent trained on a simulated or historical reward signal can learn a policy that is locally optimal but clinically unsafe in edge cases it rarely saw during training. Any recommendation must be reviewed by a licensed clinician before being acted on — the agent should assist, not replace, medical judgement.
- **Bias:** if the training data under-represents certain demographic groups (age, ethnicity, comorbidities), the learned policy may systematically under- or over-treat those groups. Regular fairness audits across subpopulations are required before and after deployment.
- **Explainability:** unlike the Decision Tree, a Q-table/RL policy can be harder to justify to a patient in plain language. Presenting the reward rationale alongside each recommendation (e.g. "Medication recommended because your risk profile is Critical") is essential for informed consent and clinician trust.
- **Reward hacking / mis-specification:** if the reward function does not perfectly capture true long-term patient wellbeing, the agent may optimise a proxy (e.g. short-term symptom reduction) at the expense of

the real goal. Reward design must be validated jointly with clinical experts, not engineers alone.

• **Accountability:** a clear human-in-the-loop governance process must define who is responsible for an adverse outcome that follows an AI recommendation — the treating physician retains ultimate clinical responsibility.